



Giardia lamblia, Cryptosporidium parvum and Balantidium coli

Niyan Inaam Muhammed Yousif

Hawler Medical University

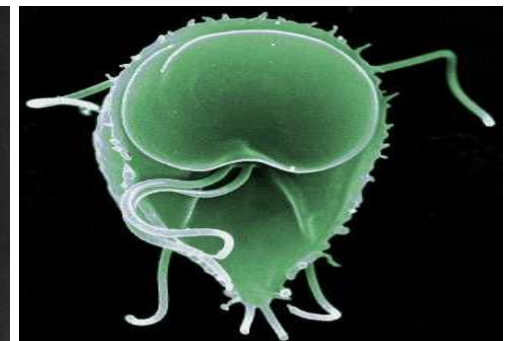
College of Medicine

Third Stage

[*niyan.inaam@hmu.edu.krd*](mailto:niyan.inaam@hmu.edu.krd)

Giardia lamblia

- **History**
- One of the earliest protozoan parasites described; observed by Antony van Leeuwenhoek in 1681.
- Giardia: Named after Professor Giard of Paris.
- Lamblia: Named after Professor Lambl of Prague, who provided a detailed description



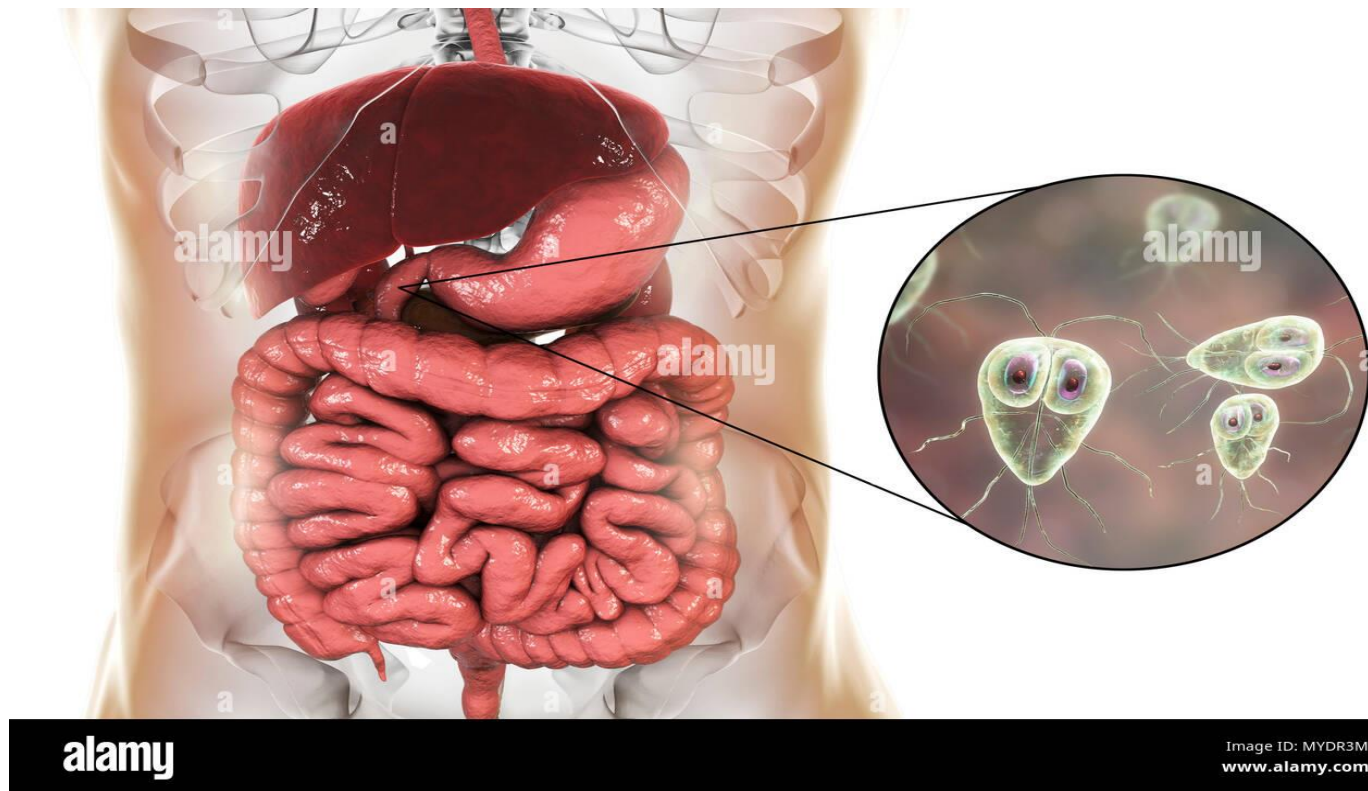
Giardia lamblia

Distribution

- One of the most common protozoan infections worldwide.
- Highly endemic in areas with poor sanitation, particularly tropical and subtropical regions.
- Common cause of traveller's diarrhoea, mainly through ingestion of contaminated water.

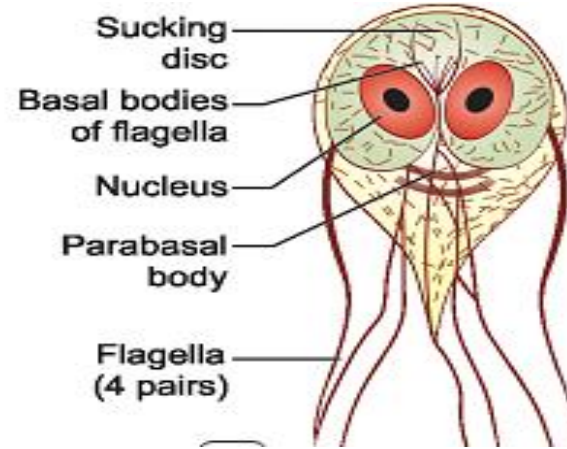
Habitat

G. lamblia lives in the duodenum and upper jejunum and is the only protozoan parasite found in the lumen of the human small intestine.

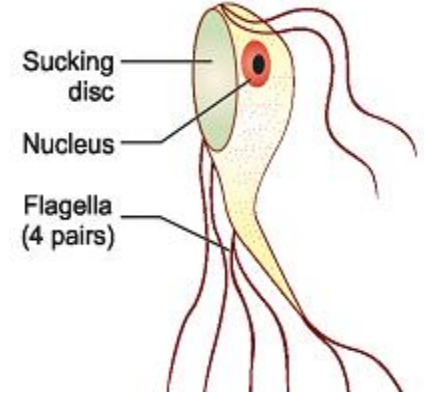


Giardia lamblia

- **Morphology**
- **Exists in two forms:**
Trophozoite and Cyst.



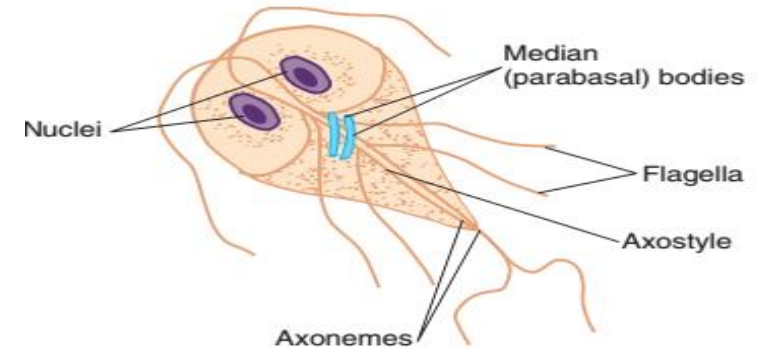
Ventral view



Lateral view

- Trophozoite: pear/heart-shaped, resembling a tennis racket measures approximately $15 \times 9 \times 4 \mu\text{m}$.
- Dorsal surface: convex; ventral surface: concave with a sucking disc for attachment to the intestinal mucosa.

Trophozoite



Size range: 8-20 μm by 5-16 μm
Average length: 10-15 μm

Bilaterally symmetrical with **2 nuclei**.

Possesses **4 pairs of flagella** for motility.

Axostyles: a pair, extending along the midline.

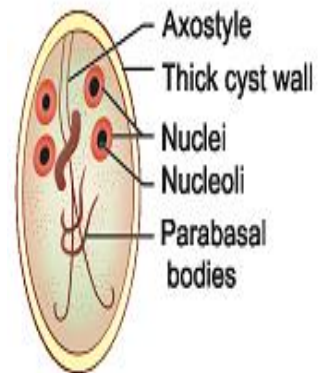
Median bodies: two sausage-shaped structures located posterior to the sucking disc.

Motility: slow, characteristic “falling leaf” movement.

Cyst



- **Infective** and environmentally **resistant form**; may survive in water for months.
- Oval-shaped, approximately $12 \times 8 \mu\text{m}$, with a hyaline cyst wall.
- Mature cyst: 4 nuclei (2 pairs); immature cyst: 2 nuclei.
- Axostyle lies diagonally within the cyst.
- Remnants of flagella and sucking disc may be present in young cysts





Mode of transmission

Giardia intestinalis found world wide in lakes, streams and other water source.

- **water borne**
- **food borne**
- **Fecal - oral transmission**

Most common intestinal parasite especially among children.

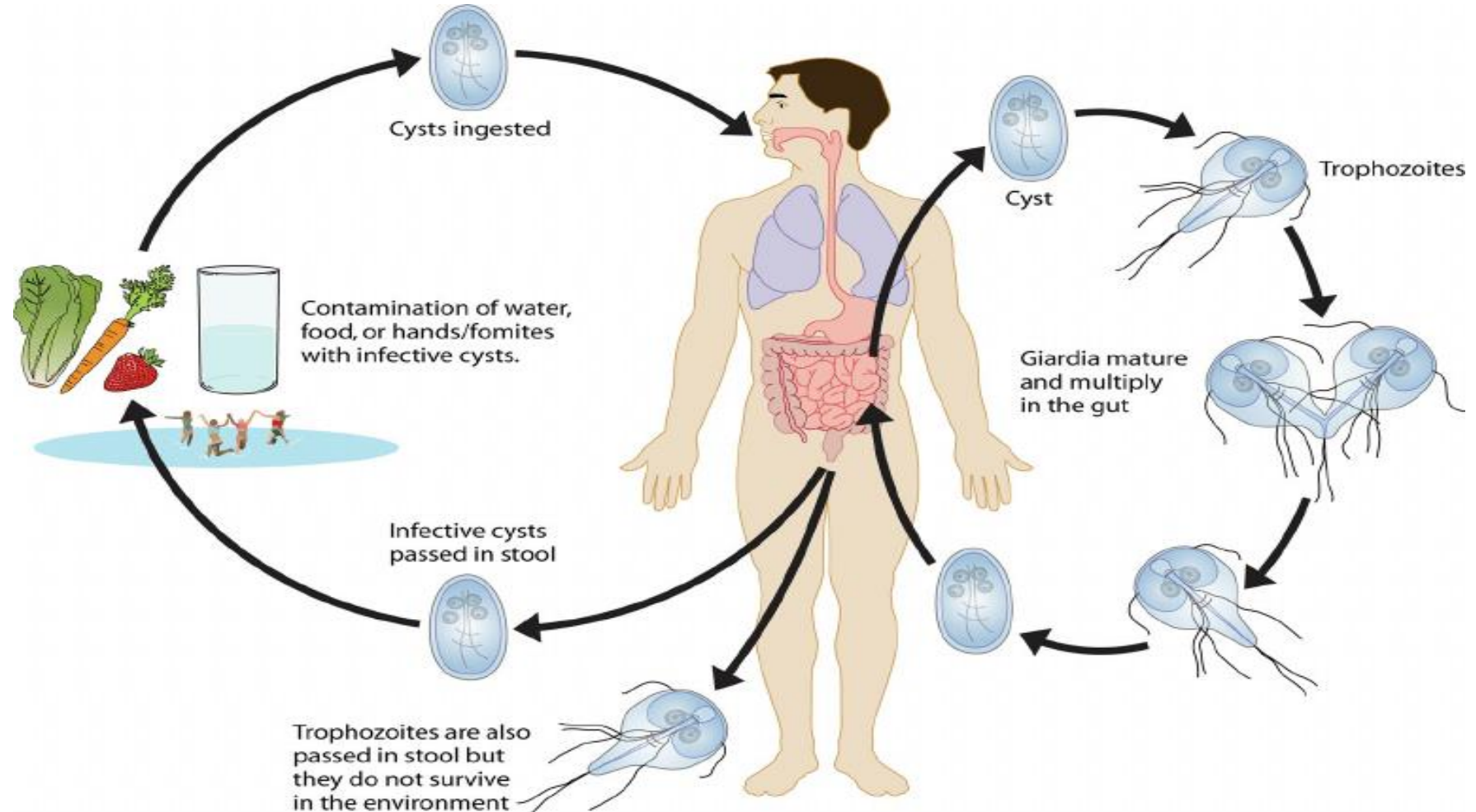
Human acquires infection by **ingestion of cysts in contaminated water and food**

Direct person to person transmission may also occur in children, male homosexuals, and mentally ill persons.

Life Cycle

Giardia passes its life cycle in **1 host**.

Infective form: **Mature cyst**.



Life Cycle

- Cysts excyst in the duodenum, releasing two trophozoites.
- Trophozoites multiply by binary fission
- Colonize the duodenum and upper jejunum.
- Cysts are passed in feces and can survive in water and soil for weeks.
- Infective dose: approximately 10–100 cysts

Pathogenesis

- Trophozoites colonize the duodenum and jejunum without tissue invasion.
- They attach firmly to intestinal epithelium via the ventral sucking disc.
- May cause villous damage, epithelial apoptosis, and lymphocytic infiltration.
- Variant-specific surface proteins (VSPs) contribute to parasite virulence and infectivity.

Clinical Features

- Often asymptomatic; symptoms usually appear after 1–2 weeks.
- Acute: **foul-smelling diarrhea**, flatulence, abdominal cramps and bloating.
- Chronic: **steatorrhea**, malabsorption, weight loss and fatigue.
- Stool may contain **mucus and fat, but usually no blood**.
- Children may develop chronic diarrhea, malabsorption and nutritional deficiencies.
- Rare extraintestinal manifestations: uveitis and retinal hemorrhage.

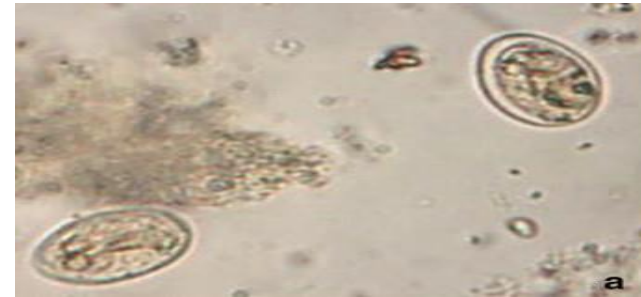


Stool Examination

Diagnosis by detecting Giardia cysts or trophozoites in stool.

Macroscopic: pale, foul-smelling, **fatty stools** that may float.

Microscopic: cysts and trophozoites identified by saline/iodine wet mount.



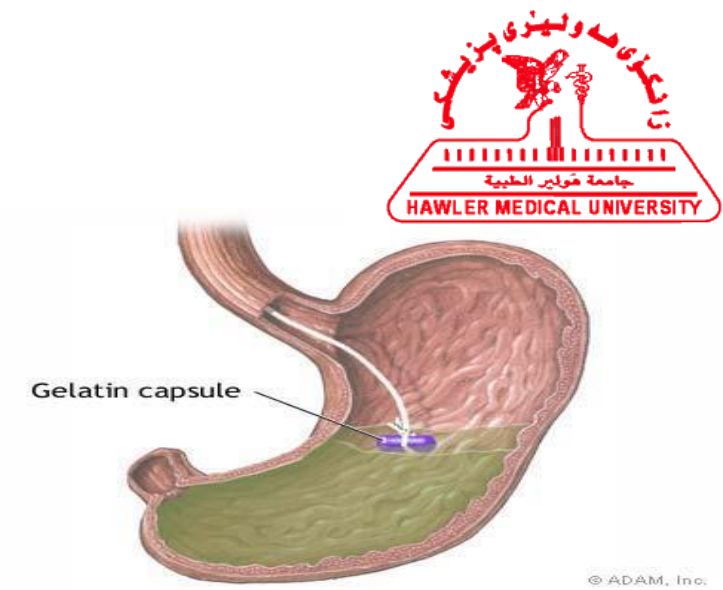
Serology:

Antigen detection: Stool ELISA/IFT; indicates active infection.

Antibody detection: ELISA/IFT; mainly useful for epidemiological studies.

Molecular diagnosis:

PCR detects Giardia DNA in stool.



Duodenal content examination:

- For detection of trophozoites , string test (entero-test).
- A useful method for obtaining duodenal specimen is entero-test.
- A swallowed gelatin capsule with an attached thread collects duodenal contents.
- The retrieved specimen is examined microscopically for trophozoites.

Treatment

- **Metronidazole or tinidazole**: commonly used first-line treatments.
- **Furazolidone or nitazoxanide**: alternatives, particularly in children.
- **Paromomycin**: an option during pregnancy, especially for symptomatic patients.

Prevention and Control

Giardiasis can be **prevented** by following measures:

- ✓ Improvement of personal hygiene.
- ✓ Prevention of food and water contamination.
- ✓ Community chlorination of water is ineffective for inactivating cysts.
- ✓ Boiling of water and filtration by filters are required.

Cryptosporidiosis

Cryptosporidiosis: parasitic infection causing **diarrhea**.

mainly due to (*Cryptosporidium parvum*).

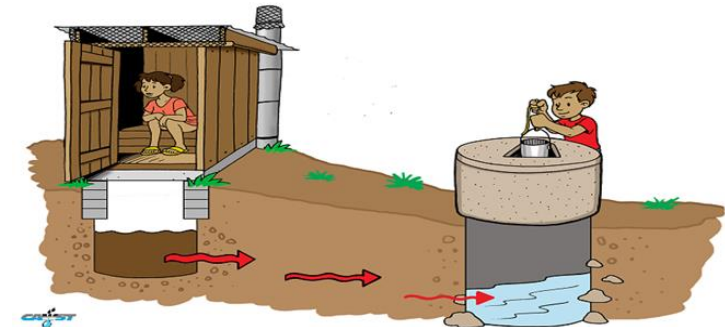


- History
- 1907: First observed by Tyzzer in laboratory mice.
- 1971: Recognized as an important cause of diarrhea in animals.
- 1976: First reported human infection.

Distribution



- Worldwide intestinal infection caused mainly by *C. hominis* and *C. parvum*.
- Usually causes self-limiting diarrhea in immunocompetent individuals.
- Fecal–oral transmission, including person-to-person spread; outbreaks occur in day-care centers and hospitals.
- An obligate intracellular apicomplexan parasite.
- Causes severe, persistent diarrhea in immunocompromised patients, especially those with AIDS.
- Most common water-borne illness in the world



Cryptosporidium parvum

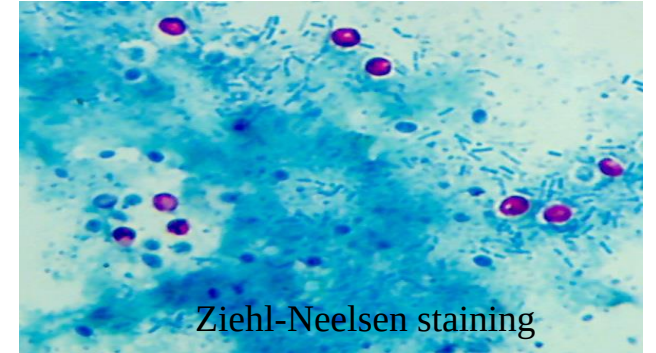
Habitat

C. parvum inhabits the small intestine. It may also be found in stomach, appendix, colon, rectum, and pulmonary tree.



Morphology

- Oocyst: infective form; spherical/oval, about 5 μm .
- Acid-fast and does not stain with iodine.
- Thick-walled oocysts: environmentally resistant.
- Thin-walled oocysts (~20%) cause autoinfection.
- Both types contain 4 crescent-shaped sporozoites.
- Highly resistant to disinfectants and environmental conditions.

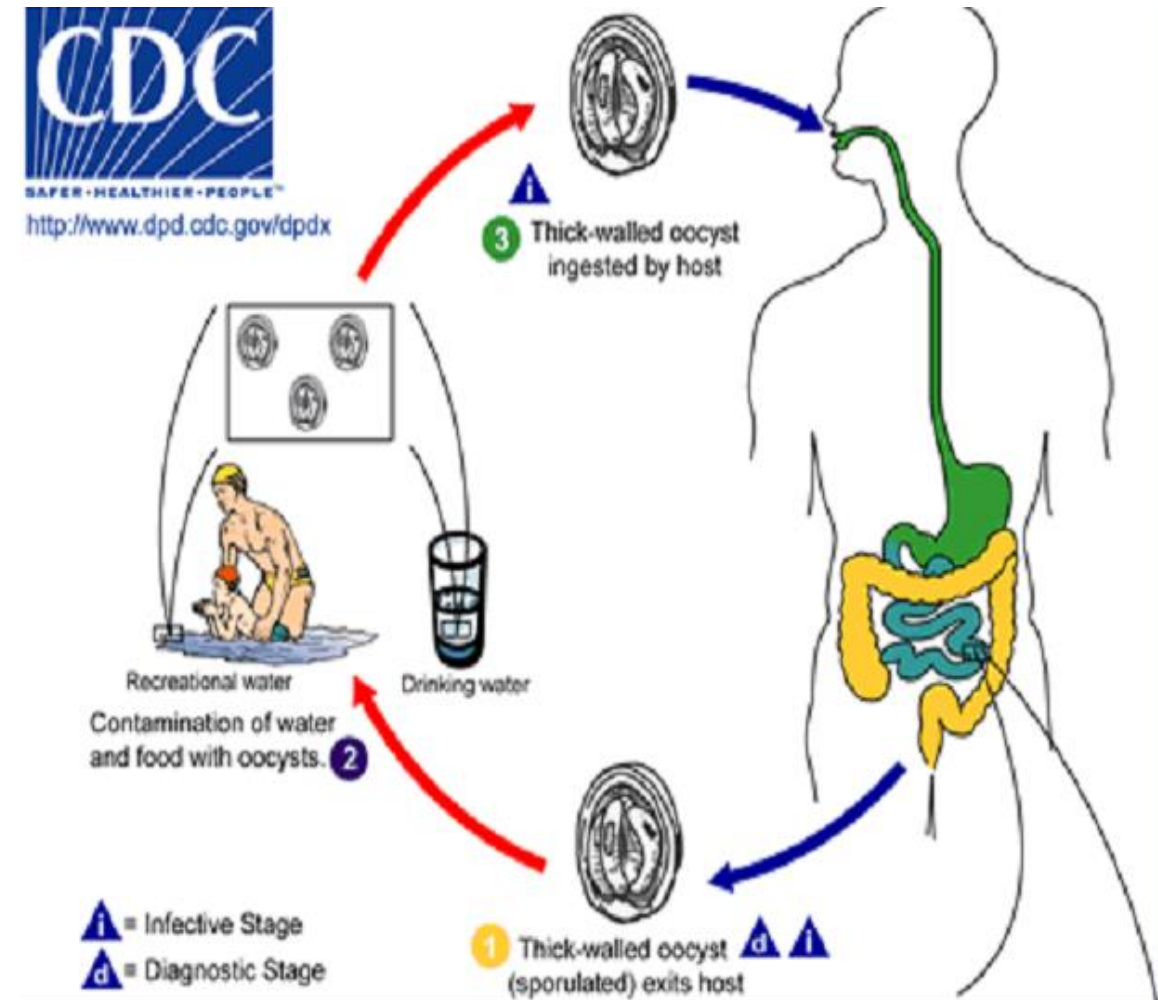


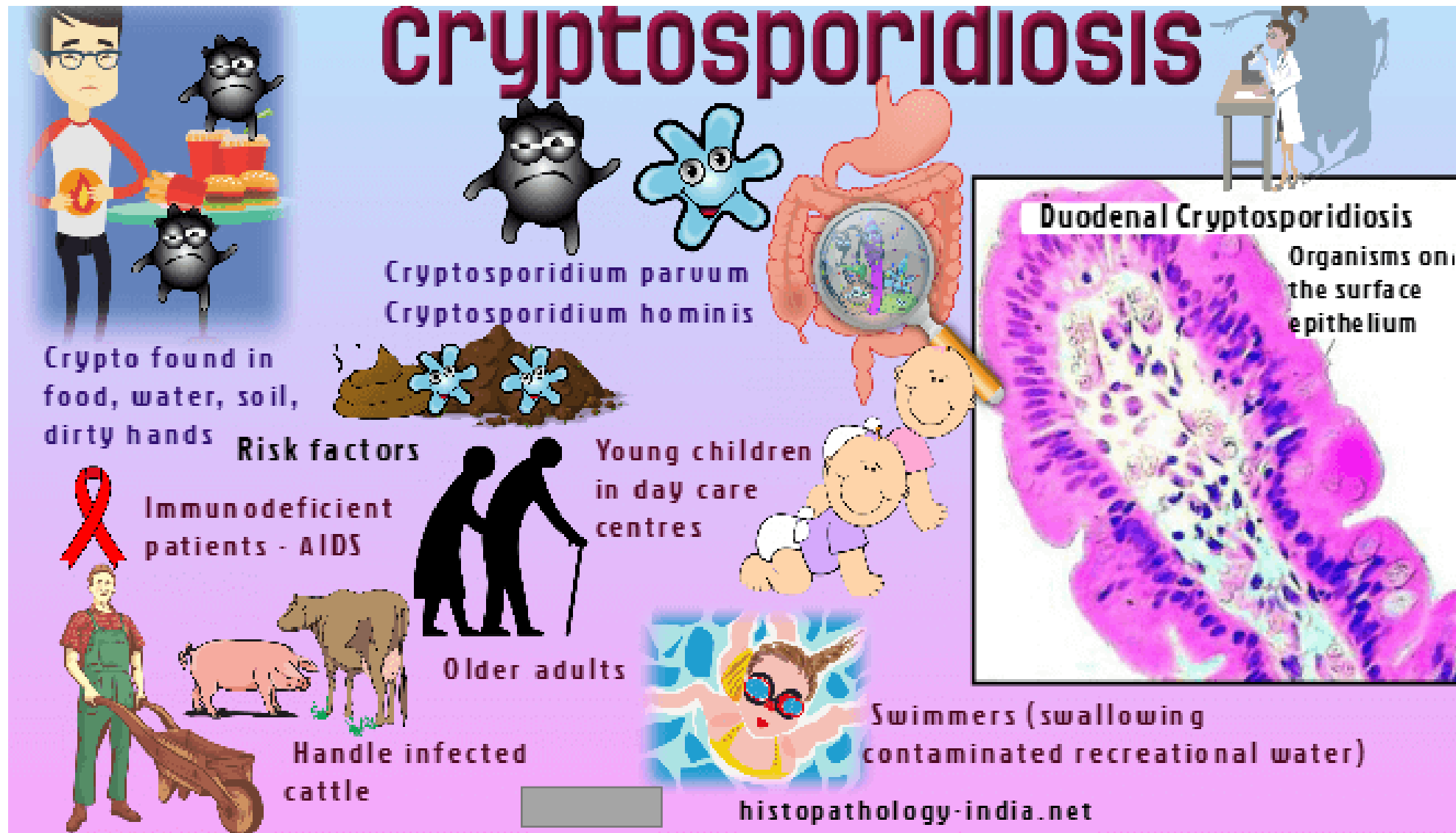
Life Cycle

The parasite complete its life cycle, sexual and asexual phases in a single host.

Host: human, birds and mammals

Infective form: Sporulated oocysts.





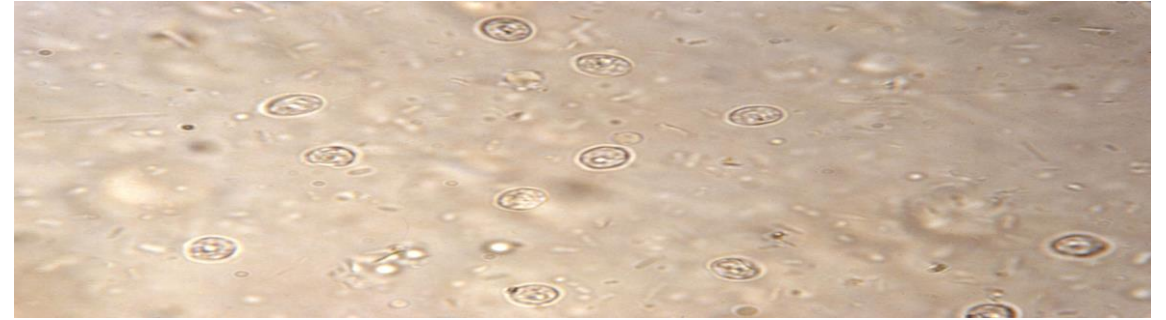
Clinical manifestations

- **Immunocompetent:** often **asymptomatic or self-limiting watery diarrhea**, with abdominal pain, nausea and weight loss.
- Children/travellers: common cause of diarrhea and waterborne outbreaks.
- **Immunocompromised (especially AIDS):** **chronic, persistent**, and remarkably **profuse diarrhea**, dehydration, electrolyte loss and marked weight loss.
- Extraintestinal: may rarely cause respiratory disease.
- Outbreaks may occur in day-care centers

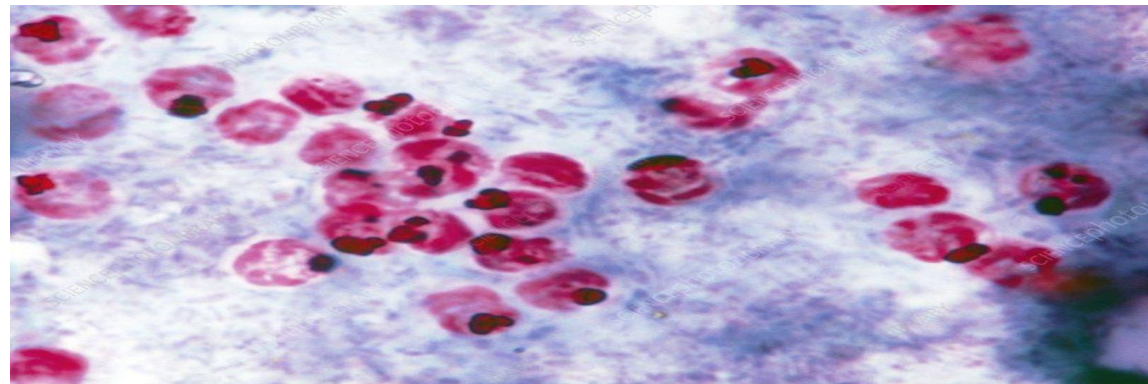
Diagnosis of Cryptosporidiosis

- Stool Examination.

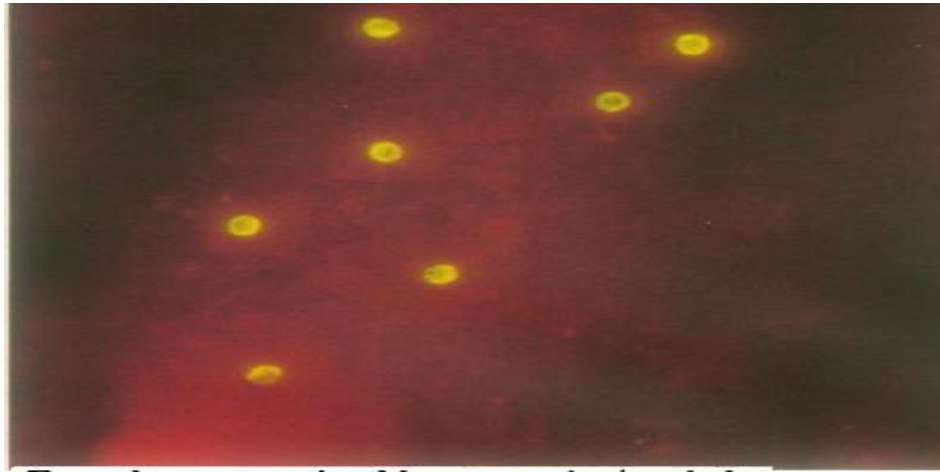
detects *Cryptosporidium* oocysts.



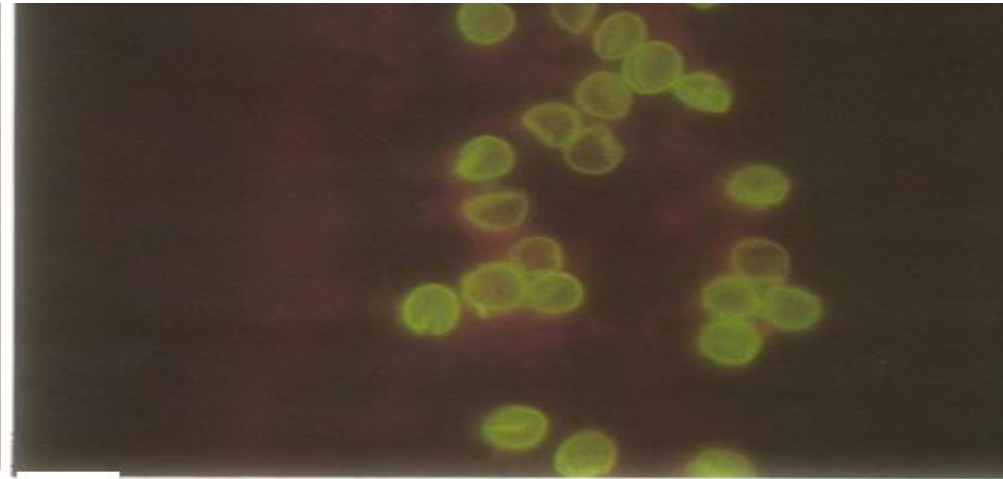
Modified acid-fast stain: method of choice; oocysts appear as red spheres.



Immunofluorescence: detects oocysts using specific antibodies



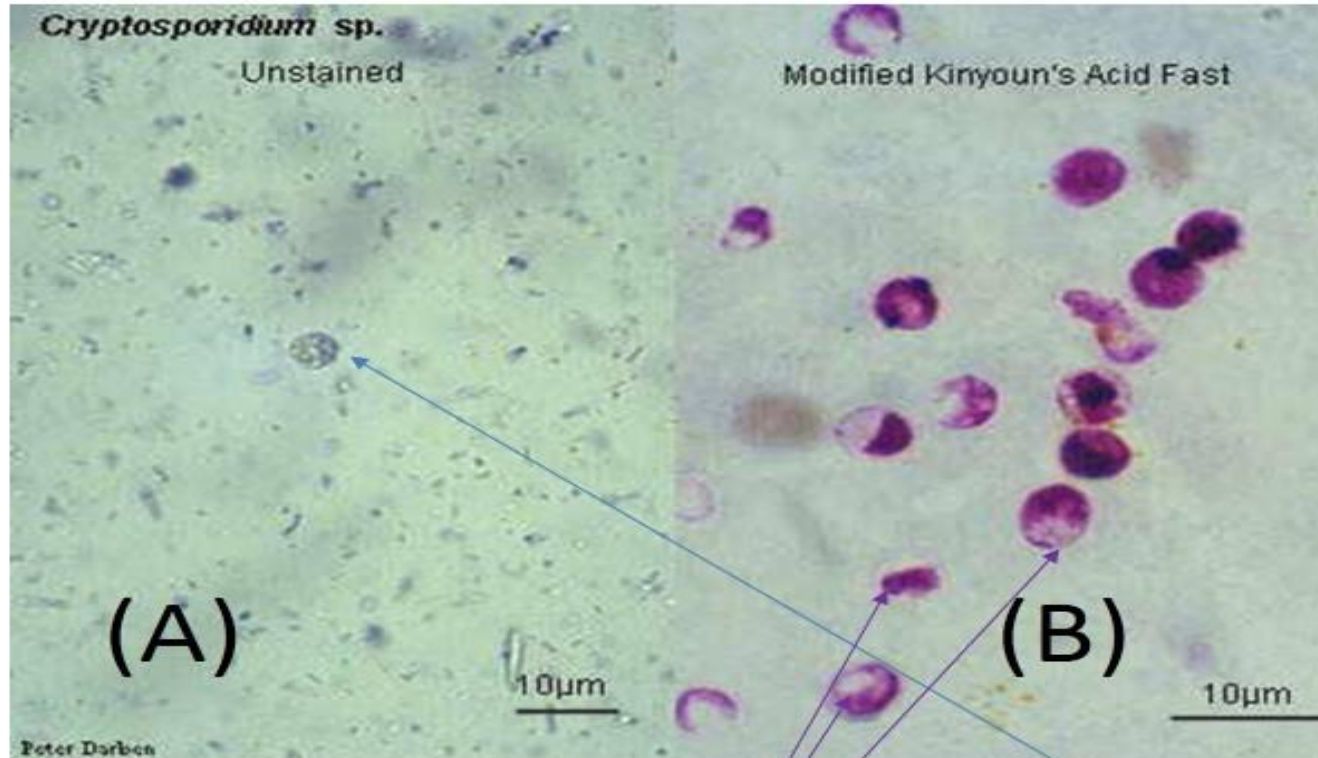
Faecal smear stained by auramine/carbol-fuchsin showing a group of oocysts with characteristic "erythrocyte" pattern of staining.



*Purified oocysts of *C. parvum* stained by monoclonal antibody conjugate, counterstained with 0.1% Evan's blue, showing the suture on the oocyst surface.*

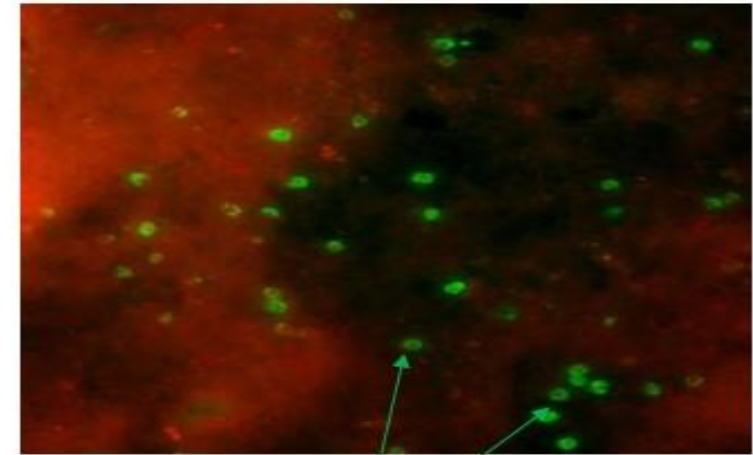
ELISA: detects *Cryptosporidium* antigen in stool.

PCR: detects parasite DNA.



(A) Unstained wet mount showing an oocyst (centre) of *Cryptosporidium* species

(B) Modified Kinyoun's acid-fast stained smear of stool showing oocysts of *Cryptosporidium* species



Direct fluorescent antibody (DFA) assay showing oocysts of *Cryptosporidium* species

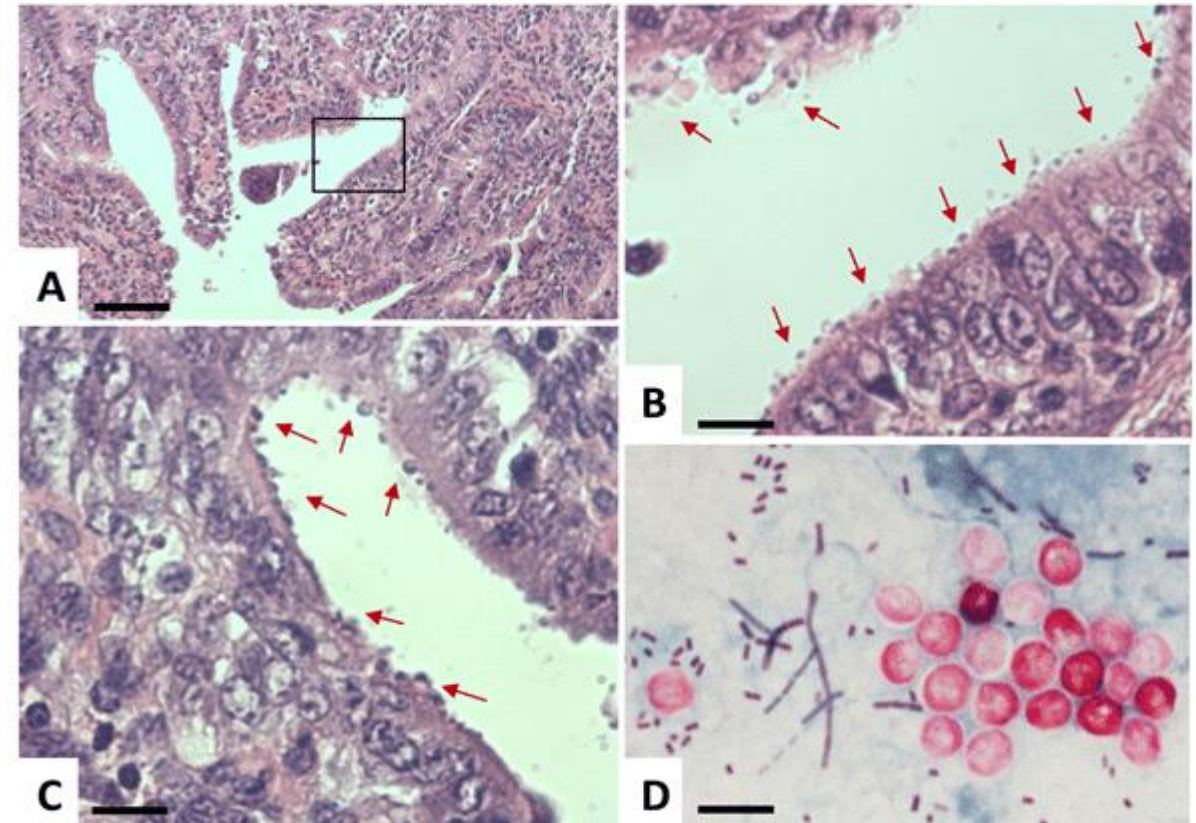
Features of this test:

- ✓ highest combination of sensitivity and specificity
- ✓ gold standard method

- **Histopathological examination (biopsy)**

Cryptosporidium can also be identified by light and electron microscope from biopsy specimen of the small bowel.

Light microscopy of *Cryptosporidium*.



(A-C) Histological sections of paraffin-embedded *C. parvum*-infected intestinal tissue, stained with hematoxylin and eosin.

(A) A low-magnification view,

(B) Arrows point towards

C. parvum parasitophorous vacuoles seen as round bodies on the surface of the epithelial layer

(D) Modified Ziehl-Neelsen staining of oocysts (red, circular bodies) in a stool sample.



Treatment



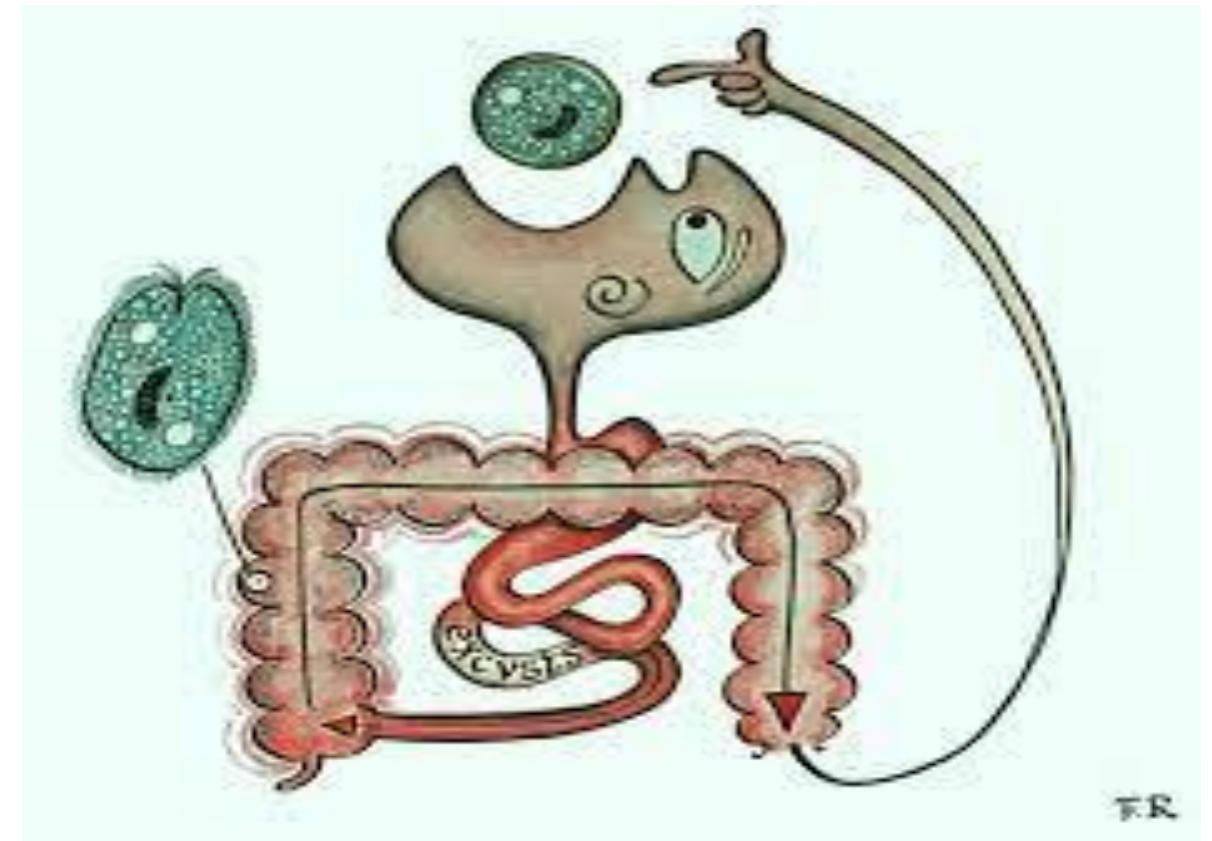
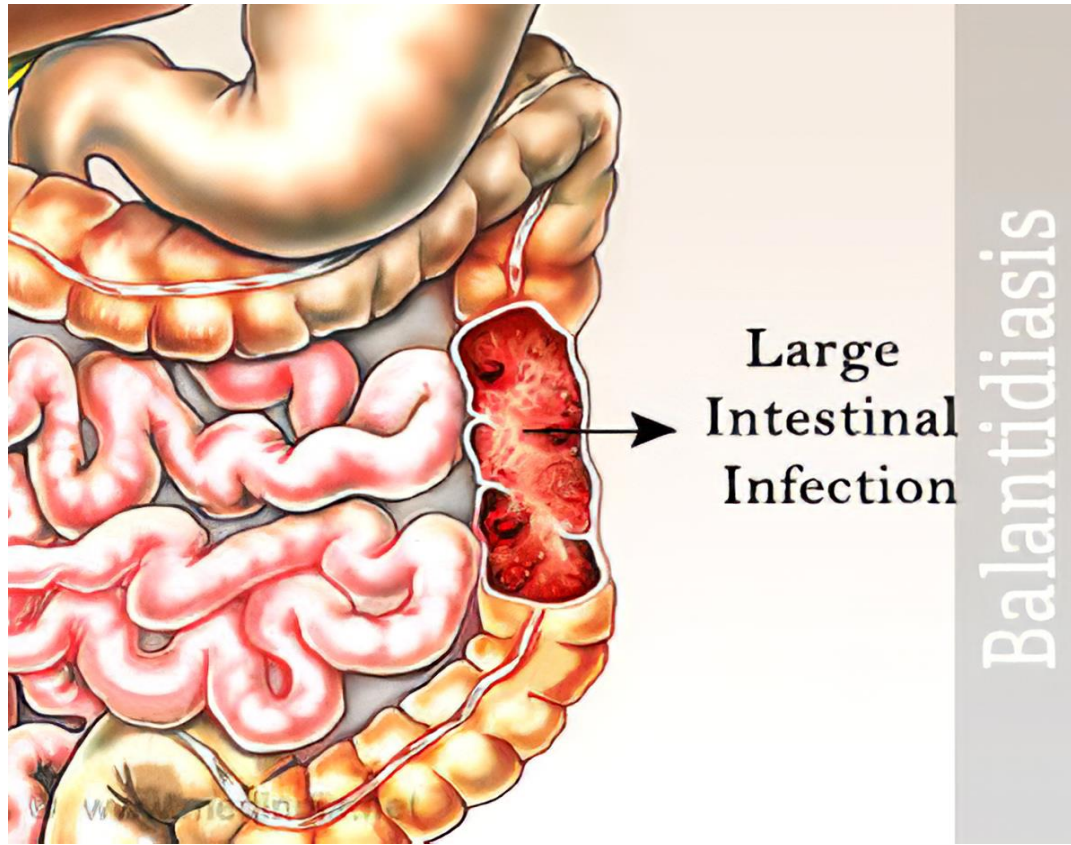
- Self limiting in most immunocompetent patients
- No effective specific therapy.
- Nitazoxanide, and supportive treatment to prove re-hydration with replacement of fluid, electrolytes and nutrients.

Prevention and control

- Improvement of personal hygiene.
- The care to avoid contamination of food and water with human and animal feces.
- Proper treatment of water supplies



Balantidiasis (*Balantidium coli*)



- History and Distribution

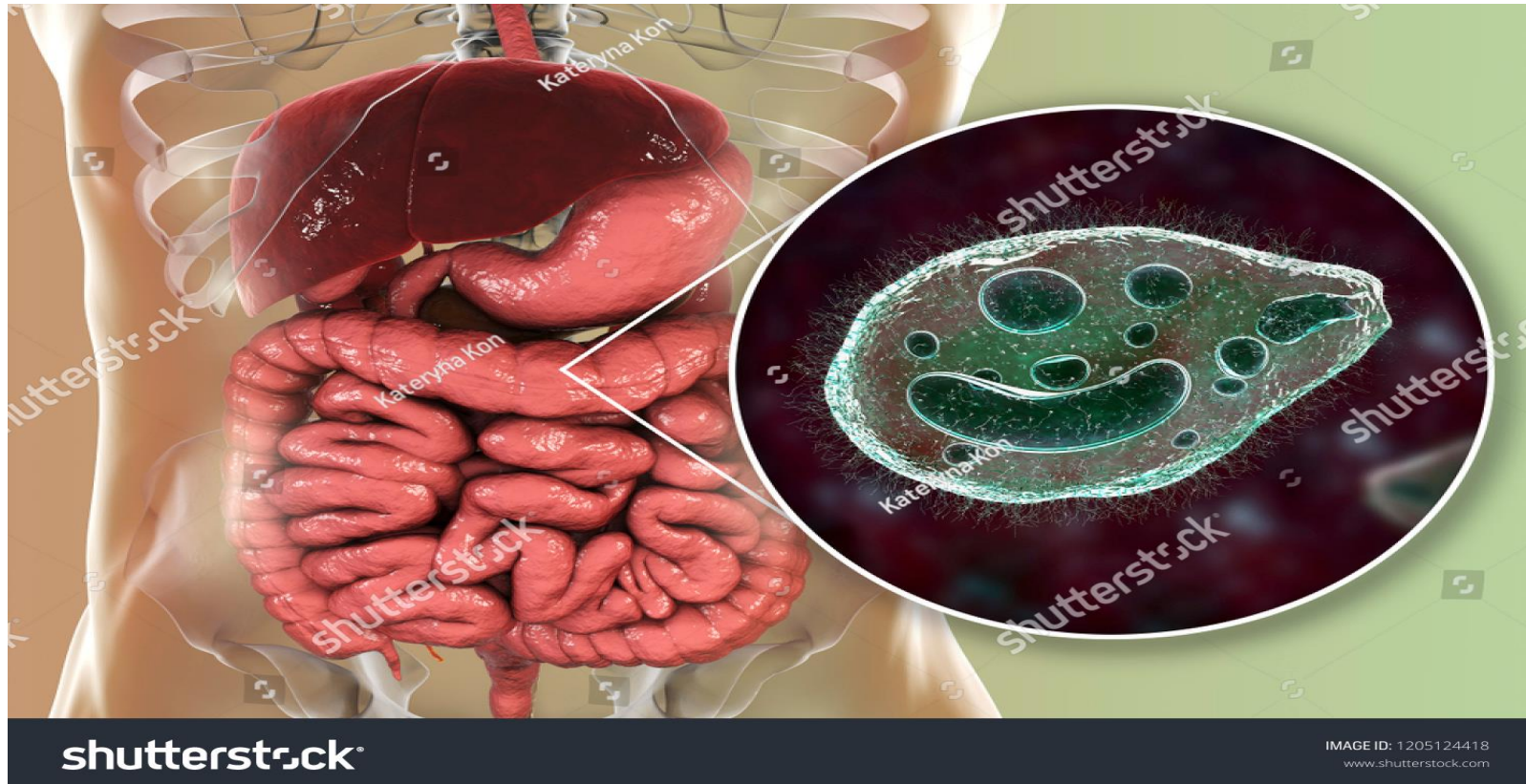
- *Balantidium coli* belongs to the phylum Ciliophora.
- The only ciliate protozoan known to infect humans.
- Largest human protozoan parasite.
- First described by Malmsten (1857) in dysenteric stool.
- Worldwide distribution, but infection is uncommon.
- Highest endemicity occurs in New Guinea, associated with close human–pig contact.



Balantidium coli

Habitat

B. coli resides in the large intestine of human, pigs, and monkeys.



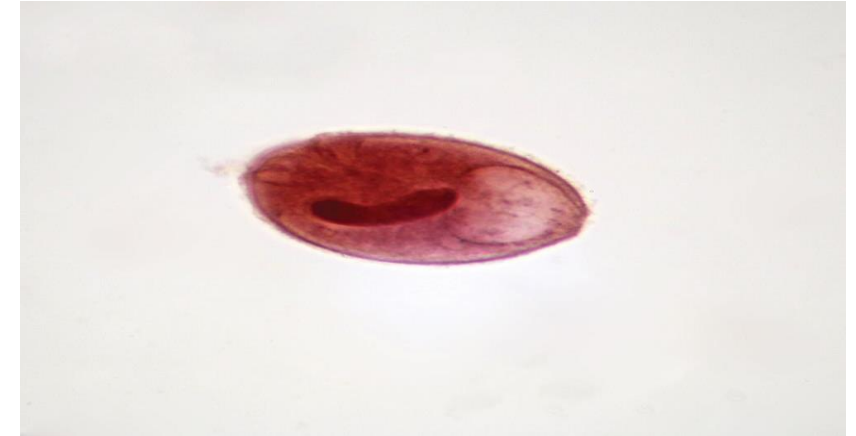
Morphology

B. Coli occurs in 2 stages :

Trophozoite and cyst

Trophozoite:

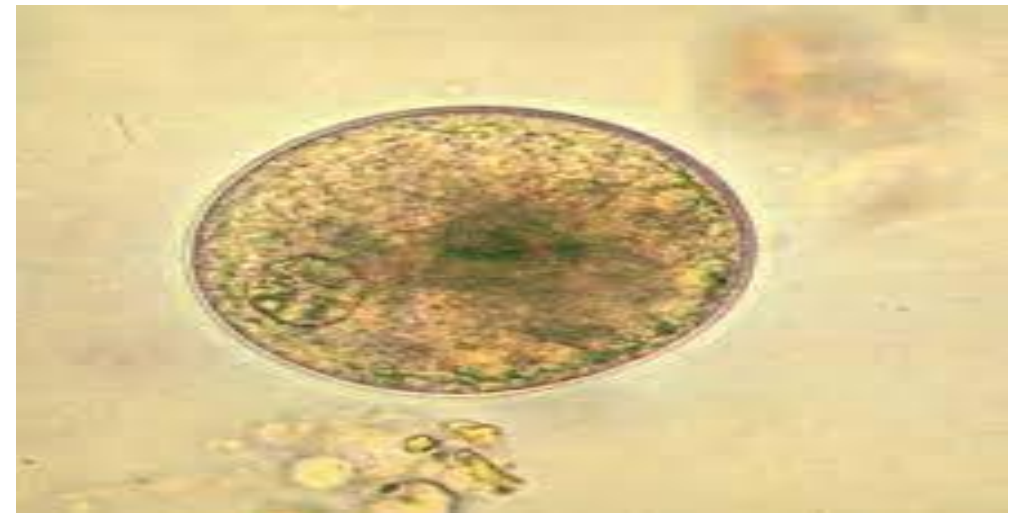
- Lives in the large intestine; 60–70 μm , ovoid.
- Has a kidney-shaped **macronucleus** and a small **micronucleus**.
- Contains food and contractile vacuoles.
- Motile and invasive stage; found in dysenteric stool.



Balantidium coli

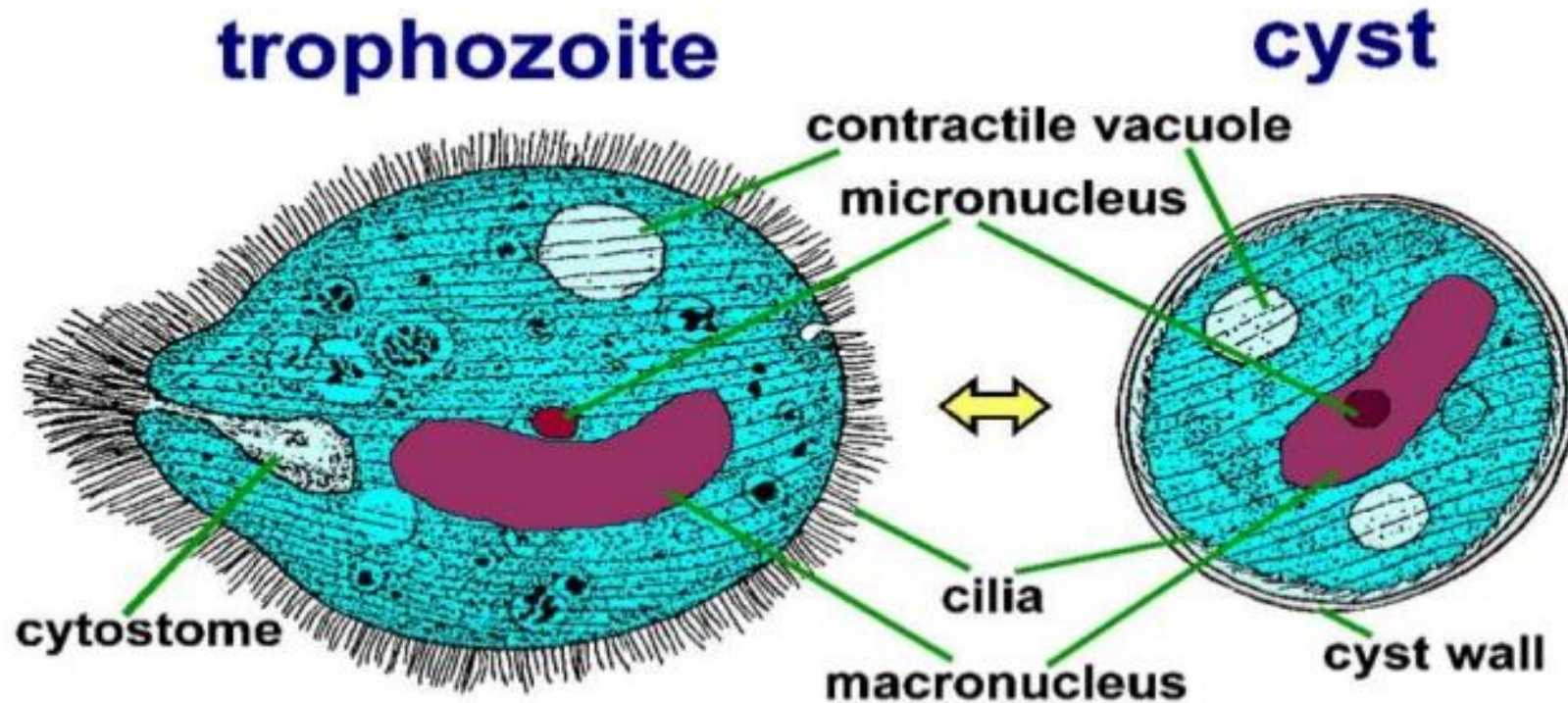
Cyst

- Spherical, 40–60 μm , with a thick double-layered wall.
- Contains macro-/micronucleus and vacuoles.
- Infective stage; common in chronic cases and carriers.



Morphology

Balantidium coli



Balantidium coli

Mode of transmission:

Balantidiasis is a zoonosis.



Human beings acquire infection by ingestion of food and water contaminated with feces containing the *cysts of B. coli*.

Infection is acquired from pigs and other animal reservoirs or from human carriers.

Life Cycle

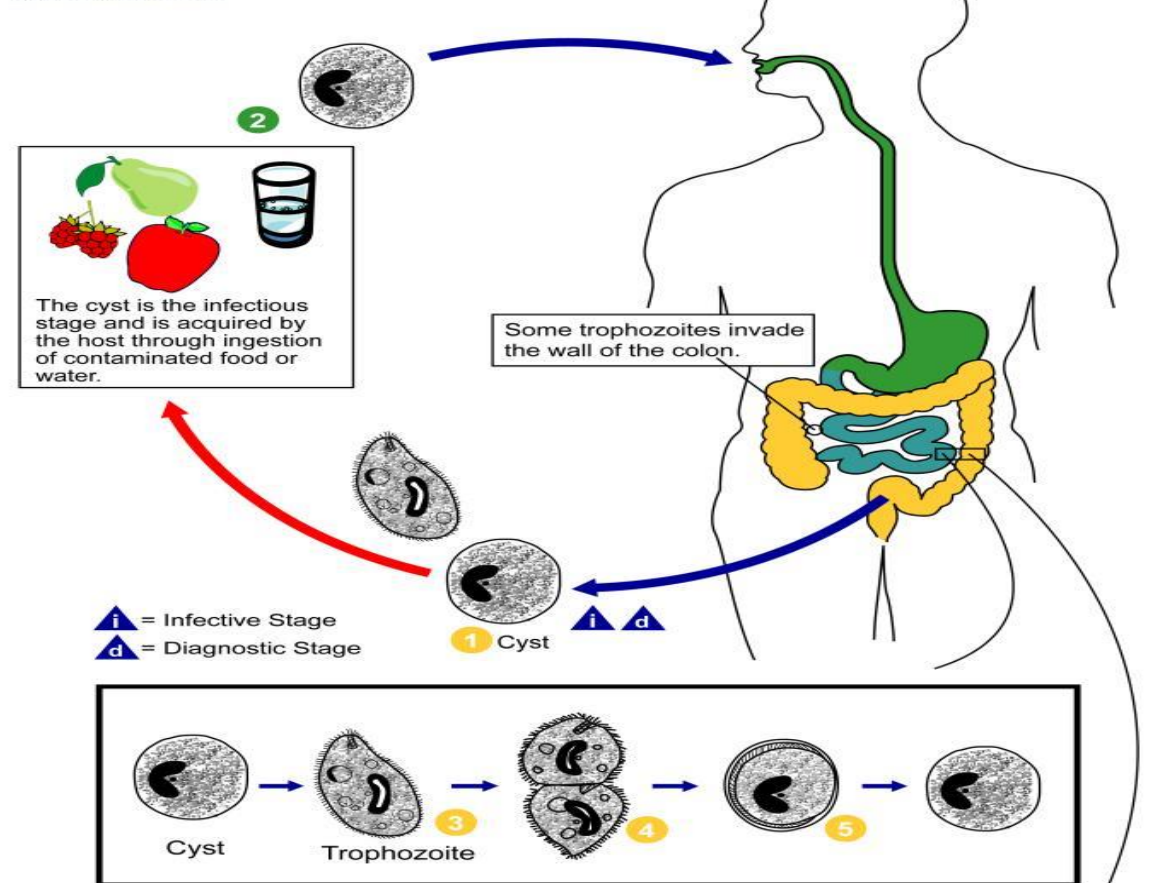
Life Cycle *B. coli* passes its life cycle in **one host** only (monoxenous).

Accidental host: Human.

Reservoirs: Pig, monkey, and rat.

Infective form: Cyst.

Balantidiasis (*Balantidium coli*)



Pathogenesis and Clinical Features

- Usually a commensal and asymptomatic infection in healthy individuals.
- Disease is more likely with malnutrition, alcoholism, or reduced host resistance.
- Trophozoites invade the colonic mucosa, causing ulcers and submucosal abscesses.
- Clinical disease resembles amoebiasis: diarrhea, dysentery, abdominal cramps, nausea and vomiting.
- Unlike *E. histolytica*, extraintestinal invasion is uncommon.

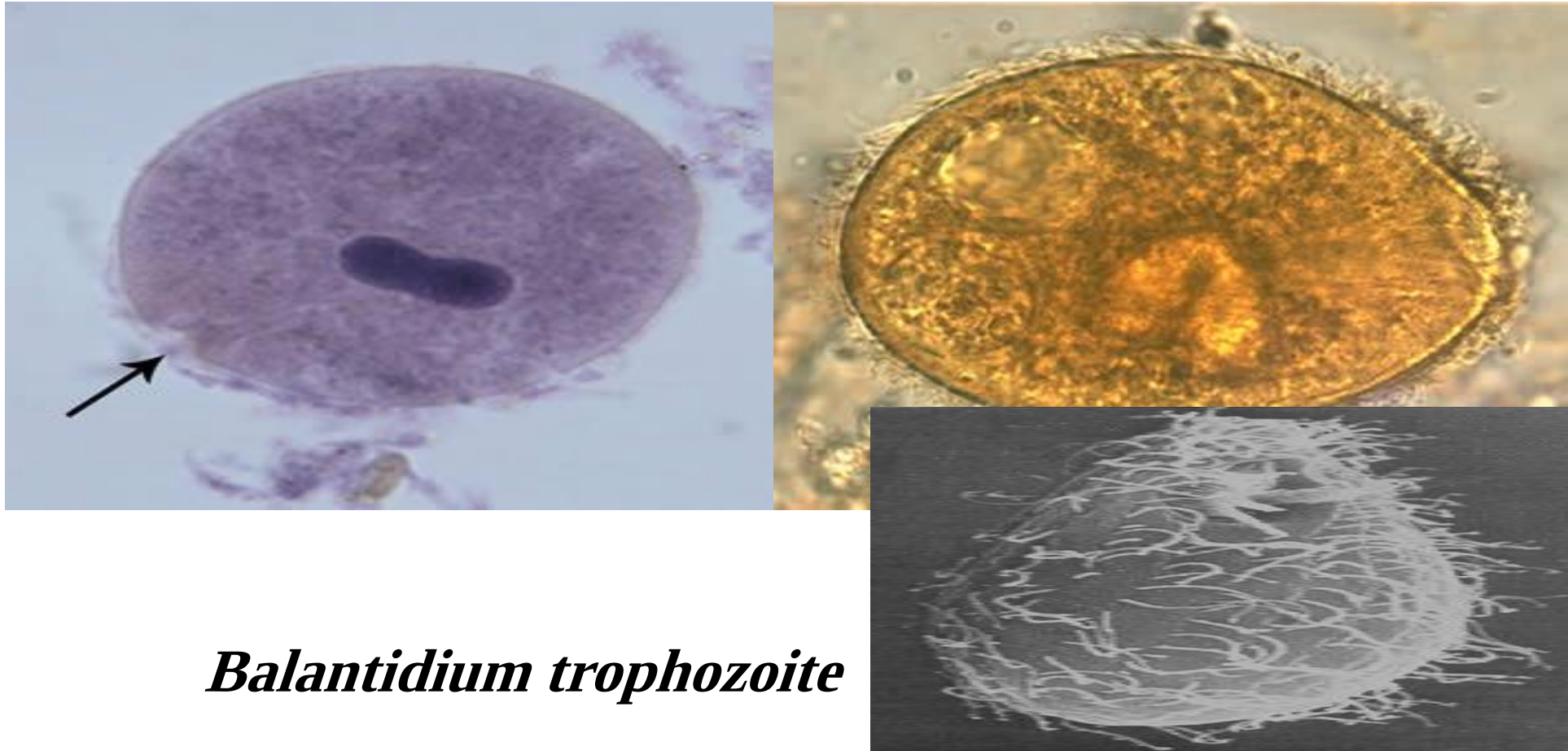
Diagnosis

Stool examination: detects trophozoites in diarrheal stools and cysts in formed stools.

Trophozoites are identified by their large size, macronucleus, and rapid motility.

Cysts are round with a prominent macronucleus.

Biopsy: intestinal ulcer tissue may be examined when stool examination is negative



Balantidium trophozoite

Treatment

Tetracycline is the drug of choice.

Alternatively Doxycycline can be give.

Metronidazole have also been reported to be useful in some cases.

Prophylaxis

Avoidance of contamination of food and water with human or animal feces.

Prevention of human-pig contact.

Improved sanitation in institutions.

Treatment of individuals shedding *B. coli* cysts

References

- ❖ Paniker CJ (2013). Paniker's textbook of medical parasitology, New Delhi, London, Philadelphia, Panama, Jaypee Brothers Medical Publishers.
- ❖ Chakarova, B., I. Miteva & S. Stanilova. 2009. Giardiasis in humans – clinical and diagnostic study. In: Proceedings of the Eighth National Conference of Parasitology, 23– 26 September 2009, Varna, Bulgaria, pp. 52–53
- ❖ Schuster, F.L and Ramirez-Avila, L. (2008). Current world status of *Balantidium coli*. Clinical Microbiology Reviews Clinical Microbiology Reviews. 21, American Society for Microbiology pp. 626-638.
- ❖ Ryan U, Fayer R, Xiao L. *Cryptosporidium* species in humans and animals: current understanding and research needs. Cambridge University Press. 2014; 141: 1667–1685. doi: 10.1017/S0031182014001085
- ❖ CDC for figures and illustrations.